

[Language Translators]/ [Third]

Now we must Create an array of record to save the output of the lexical step instead of print it on the output screen as follows: -



[Lab 4]

[08:30 Am]

1- Create * Class Record → have 2 var (lexem ,type) as String
Initialize this record

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*create array of this record

*define the index of array=0

Hours

[2 Hours]

```
public static class Record {
    String lexem;
    String type;

    Record (String L,String T)//initialization
    {
        lexem =L;
        type=T;
    }
    }// End of record
public static Record[] Lut=new Record [100];
public static int Lutn=0;
```

2- Create methods to insert data in this array after checking the type of token: -

*Insert_lut → (RW,Dig,ID)

*Insert_ch →(Sp,Op)

```
public static void insert_Lut(String tok){
    if(is_RW(tok)){Lut[Lutn]=new Record(tok,"RW");Lutn++;}
    elseif(is_DIG(tok)){Lut[Lutn]=new Record(tok,"DIG");Lutn++;}
    else if (is_ID(tok)) {Lut[Lutn]=new Record(tok,"ID");Lutn++;}
    else System.out.println(tok+" it is Error");
}

public static void insert_ch(char c){
    if(is_OP(c)){Lut[Lutn]=new Record(String.valueOf(c),"OP");Lutn++;}
    elseif(is_ (c)){Lut[Lutn]=newRecord(String.valueOf(c),"SP");Lutn++;}
}
```

3- Now call the insertion method in main method instead of the previous code (to optimize our code).*

4- After finishing the lexical work, we must close the text file and print the output (array of record).*

```
public static void main(String[] args) throws Exception{
    Scanner in;
    /*read from txt file*/
```

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File ("C:\Users\Khyro\Desktop\nt.txt"));

[Lab 4]

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Hours

[2 Hours]

[08:30 Am]

```

    while(in.hasNext()){
        line = in.nextLine();/*read one line from file*/
        tk = "";
        for(int i=0; i<line.length(); i++){
            ch = line.charAt(i);
            if (ch!='\n')
            {
                if(is_SP(ch)||is_OP(ch)){/* must be send to
LUT*/
                    if (!"".equals(tk)){
                        insert_Lut(tk);
                        tk="";
                        insert_ch(ch);
                    }
                    else
                        tk+=ch;
                }
                else if (!"".equals(tk)){
                    insert_Lut(tk);
                    tk="";
                }
            }
        }/*end for loop*/
        if (!"".equals(tk)) { insert_Lut(tk); tk=""; }
    }/*end while loop*/
    in.close(); /*close text file*/
    for (int j=0;j<=Lutn-1;j++)
        System.out.println(Lut[j].lexem+"      type is
"+Lut[j].type);
    }/*end main1 method*/

}/*end Main class*/

```